

Euclid's Elements — Book I

Proposition 7

Formal Reconstruction — Technical Documentation

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Uniqueness from two prescribed distances

1 Introduction

Proposition I.7 is a uniqueness theorem. A fixed base AB and two prescribed distances from its endpoints determine at most one vertex on a chosen side of the base. In modern synthetic language, if C and D lie on the same side of the line AB and

$$AC \cong AD, \quad BC \cong BD,$$

then $C = D$.

The proposition is historically important because Euclid uses it immediately in Proposition I.8: after superposition, I.7 rules out a second position of the remaining vertex. The current Lean development reverses that logical order. The general theorem `HilbertSSS` already exists, so I.7 is proved by SSS followed by two uniqueness principles: uniqueness of the ray constructed by a prescribed angle, and uniqueness of a point at a prescribed distance on a ray.

The audit therefore separates three layers:

1. Euclid's representative classical contradiction;
2. the actual proposition-level Lean proof;
3. reusable lower theorems, especially `HilbertSSS`, whose internal SSS engine has already been exposed in the audit of I.8.

2 The Classical Configuration

Let A, B be distinct points on a fixed line and let C, D be points on the same side of that line. Assume

$$AC \cong AD, \quad BC \cong BD.$$

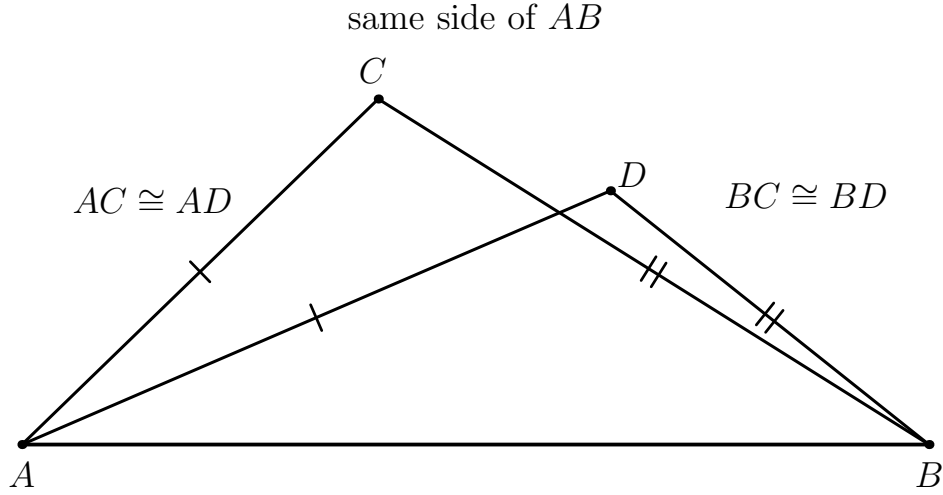


Figure 1: The forbidden hypothesis of Proposition I.7. The congruence marks encode the assumptions $AC \cong AD$ and $BC \cong BD$ while C, D are drawn as distinct points on the same side of AB . Since the theorem says that such a metric configuration cannot exist, the drawing is a reductio schematic rather than a metrically realizable Euclidean model.

The reductio hypothesis is $C \neq D$.

Euclid joins C to D . This creates two isosceles triangles, ACD and BCD , sharing the side CD . In the representative positional case the angle-order relation visible after this join conflicts with the two isosceles base-angle equalities.

3 Statement

Theorem 1 (Euclid I.7). *Let A, B be distinct points on a line base, and let C, D lie on the same side of base. If*

$$AC \cong AD, \quad BC \cong BD,$$

then

$$C = D.$$

The current production theorem records the carrier line explicitly. Besides $A \neq B$, it assumes that A and B lie on **base**, and packages the positional hypothesis as **HilbertSameSide**, applied to Geo, C, D and **base**. The target is the point equality $C = D$.

Diagrammatic audit. The preserved Book1R chapter already improved the old diagram set by merging two duplicate introductory pictures. The two resulting classical figures are retained: the forbidden hypothesis and Euclid’s joined configuration.

The old formal material, however, was only a theorem-call flowchart. The current audit replaces it by Figure 3, which records the actual proposition-level geometry in two panels. Panel (a) is the stable state after SSS: the two candidate rays from B make congruent angles with the common base ray and lie on the prescribed side. Panel (b) shows what the two uniqueness theorems accomplish: Hilbert III.4 identifies a single ray carrier through a witness X , and segment-construction uniqueness then collapses the two equal-distance candidates to $C = D$.

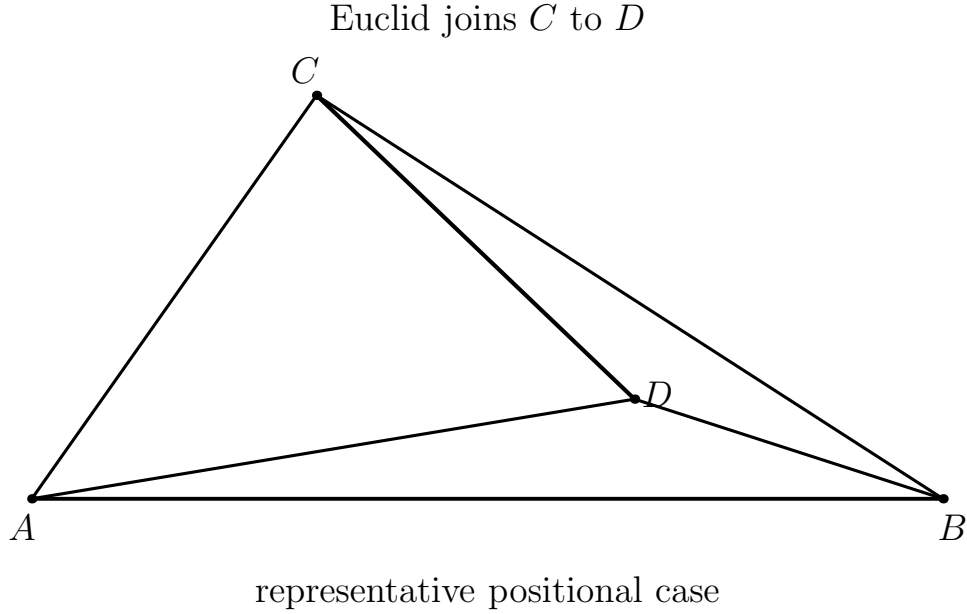


Figure 2: Euclid’s representative configuration after joining C to D . The picture records one positional case of the classical contradiction; it is not intended to encode every possible same-side position of two distinct candidate vertices.

The internal construction used to prove **HilbertSSS** is not redrawn here, because it is a reusable lower theorem and was audited explicitly in Proposition I.8. Thus the final set is

$$2 \text{ classical panels} + 2 \text{ formal panels} = 4 \text{ panels in 3 figures.}$$

4 Classical Proof

Assume $C \neq D$ and join C to D . From

$$AC \cong AD$$

triangle ACD is isosceles, so Proposition I.5 gives

$$\angle ACD \cong \angle CDA.$$

Similarly,

$$BC \cong BD$$

makes BCD isosceles, hence again by I.5,

$$\angle BCD \cong \angle CDB.$$

In Euclid’s displayed positional case, one of the latter angles is a proper part of the corresponding former angle. The two equalities above therefore force an angle to be both equal to and strictly greater than the corresponding angle, a contradiction. Euclid concludes that the two candidate vertices cannot be distinct.

A diagrammatic qualification is essential. The single classical drawing does not exhaust all possible same-side positions of C and D ; a fully expanded classical treatment requires additional

positional cases. The formal theorem below does not inherit this limitation: its `HilbertSameSide` hypothesis covers the full same-side statement.

The classical proposition dependency relevant to Euclid’s argument is

$$\boxed{\text{I.5} \longrightarrow \text{I.7}}.$$

The previous documentation listed Proposition I.6 as part of the classical route. That edge is removed in this audit: the displayed I.7 contradiction uses the isosceles base-angle theorem I.5, not the converse theorem I.6.

5 Proof Architecture of the Reconstruction

The public source begins with a trivial equality split:

$$C = D \quad \text{or} \quad C \neq D.$$

The first branch is immediate and carries no geometric content. In the second branch, the actual proposition-level DAG is

$$\begin{array}{c} C, D \text{ on the same side of } AB, \quad AC \cong AD, \quad BC \cong BD \\ \downarrow \\ C, D \text{ off the base line} \\ \downarrow \\ ABC, ABD \text{ noncollinear} \\ \downarrow \\ AB \cong AB \text{ and the two given side congruences} \\ \downarrow \text{HilbertSSS} \\ \angle ABC \cong \angle ABD \\ \downarrow \text{hilbert_angle_unique_common_ray} \\ \exists X : \text{SameRay}(B; X, C) \wedge \text{SameRay}(B; X, D) \\ \downarrow \text{hilbert_segment_construction_unique} \\ C = D. \end{array}$$

This is not Euclid’s angle-order contradiction. The formal proof turns the problem into a rigidity chain:

$$\text{SSS} \implies \text{unique ray} \implies \text{unique point on that ray}.$$

6 Hilbert Reconstruction

6.1 Off-line information and nondegeneracy

The same-side hypothesis already contains the information that both C and D are off the base line. The proof extracts those two facts and calls `hilbert_not_collinear_of_off_line` twice to obtain

$$\neg \text{Collinear}(A, B, C), \quad \neg \text{Collinear}(A, B, D).$$

These are the nondegeneracy obligations needed for the congruence step and for the subsequent angle-uniqueness theorem.

6.2 SSS creates equal angles at the common base vertex

The triangles ABC and ABD share AB . Together with

$$AC \cong AD, \quad BC \cong BD, \quad AB \cong AB,$$

the reusable theorem `HilbertSSS` gives complete triangle congruence data. I.7 consumes only the angle at B :

$$\angle ABC \cong \angle ABD.$$

At this point the formal proof has already left Euclid's historical order: SSS is available as lower infrastructure, while Euclid proves I.7 precisely in order to use it later inside his proof of I.8.

6.3 Hilbert III.4 forces a common ray

The current helper `hilbert_angle_unique_common_ray` is not merely a name for the conclusion. Its proof invokes `HilbertCongruence.angle_construction`. Applied here with common vertex B , reference ray BA , and the prescribed side of the base, it constructs a witness X for the unique ray making the required angle. The uniqueness field returned by Hilbert III.4 is then applied once to C and once to D . The result is

$$\text{SameRay}(B; X, C), \quad \text{SameRay}(B; X, D).$$

Thus the two initially distinct candidate directions have been identified at the ray level.

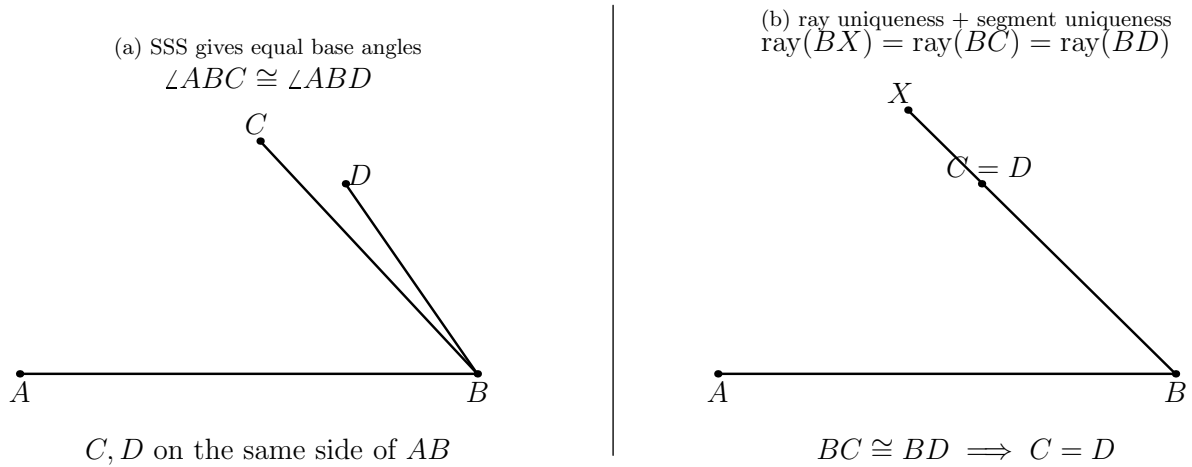


Figure 3: The proposition-level Hilbert mechanism. (a) SSS yields $\angle ABC \cong \angle ABD$ while C, D are on the prescribed side of the base. Hilbert III.4 uniqueness therefore identifies both candidate rays with one ray represented by BX . (b) Once the ray is common, the original congruence $BC \cong BD$ and uniqueness of segment construction on that ray leave only one endpoint, so $C = D$.

6.4 Segment uniqueness identifies the endpoint

The final step uses the original equality of distances from B . The proof first reverses the congruence orientation to obtain

$$BD \cong BC.$$

Both C and D lie on the ray represented by BX . The theorem `hilbert_segment_construction_unique` therefore says that two points laid off on that same ray at the same prescribed distance from B must be the same point. Hence $C = D$.

The witness X has no independent classical significance. It is a carrier witness for the unique ray supplied by the formal angle-construction API. This is why it is shown only in the formal figure and not introduced into the classical diagrams.

7 Lean Formalization

The current production proof applies SSS as follows:

```
have hSSS :=
  HilbertSSS
  Geo
  A B C
  A B D
  hABC
  hABAB
  hBCBD
  hACAD
```

It extracts the angle at B and feeds it to the current same-ray uniqueness helper:

```
rcases
  hilbert_angle_unique_common_ray
  Geo
  A B C D
  base
  hAB
  hAbase
  hBbase
  hCbase
  hSame
  hAngleB with
  <X, hRayXC, hRayXD>
```

Finally, the two points are identified by segment-construction uniqueness:

```
exact
  hilbert_segment_construction_unique
  Geo
  B C
  B X
  C D
  hRayXC
  hRayXD
  (hilbert_congruent_reflexive Geo B C)
  hBD_BC
```

The angle helper itself currently has the proposition-level specification

$$\begin{aligned} \angle OLH \cong \angle OLK, \quad H, K \text{ on the same side of } OL \\ \implies \exists X \text{ (SameRay}(L; X, H) \wedge \text{SameRay}(L; X, K)). \end{aligned}$$

Its implementation uses the uniqueness component of Hilbert III.4 rather than an angle measure or coordinate calculation.

8 Relationship with Earlier and Later Propositions

8.1 Proposition I.5

I.5 is the decisive proposition in Euclid’s displayed contradiction. The equalities $AC \cong AD$ and $BC \cong BD$ produce two isosceles triangles, and I.5 converts each side equality into a base-angle equality.

8.2 Proposition I.6 is not a dependency

The older chapter described I.5 and I.6 as a two-way bridge used in I.7. That overstates the classical proof. I.6 is not invoked in the displayed argument of Proposition I.7, and the current Lean theorem does not call `euclid_proposition_6`. It is therefore removed from both the historical and formal DAGs of this proposition.

8.3 Transition to Proposition I.8

The historical direction is

$$\text{I.7} \longrightarrow \text{I.8}.$$

After superposition in I.8, the remaining vertex is characterized by its two distances from the endpoints of the common base; I.7 excludes a second same-side realization.

The current formal library has the reverse dependency pattern at the reusable-theorem level:

$$\text{HilbertSSS} \longrightarrow \text{euclid_proposition_7}.$$

This is not circularity. The Hilbert theorem is established independently of Book I’s proposition order, and the Book I endpoint is reconstructed from that stronger infrastructure.

9 Hilbert Components Used

The proposition-level proof uses the following established components:

- `HilbertSameSide` for the prescribed half-plane and off-line information;
- `hilbert_not_collinear_of_off_line` for triangle nondegeneracy;
- `HilbertSSS` for the angle congruence at B ;
- `hilbert_angle_unique_common_ray`, whose current proof uses Hilbert III.4 angle construction and its uniqueness field;
- `hilbert_segment_construction_unique` for the final endpoint uniqueness on the common ray;
- reflexivity, symmetry, and endpoint transport for segment congruence.

No numerical distance, angle measure, coordinate system, or Euclidean parallel axiom occurs in the proposition-level argument.

10 Formalization Notes

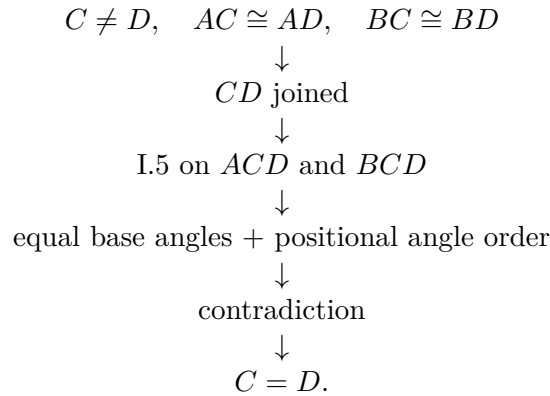
The proof has a useful modular structure. Its only local case split is whether $C = D$. The equality branch closes immediately; the distinct branch contains all of the geometry. There is no positional subdivision analogous to the classical region analysis.

The SSS call should be read as a reusable lower theorem, not as an unexplained primitive. Its internal Hilbert construction has already been audited in I.8, so recursively reproducing that construction here would obscure rather than clarify I.7's own contribution.

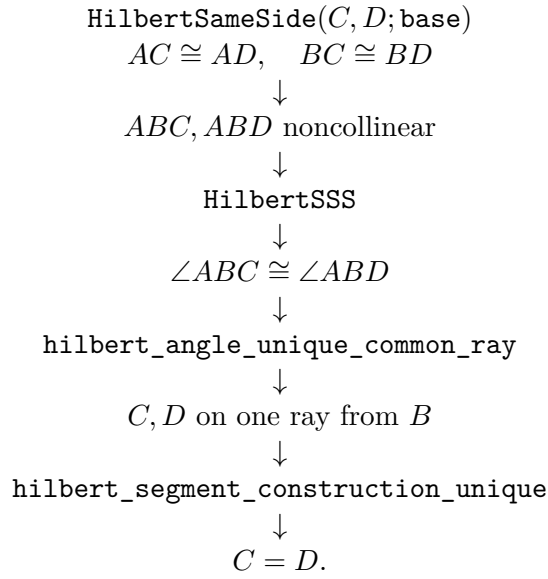
Conversely, the old theorem-call flowchart was too compressed because it hid the geometric meaning of the angle-uniqueness step. The new formal figure therefore exposes the one new stable state that matters locally: two same-side equal-angle candidates are forced onto one ray, after which segment uniqueness finishes the proof.

11 Dependency Summary

Classical representative route:



Current formal route:



12 Conclusion

Proposition I.7 is a rigidity theorem whose classical and formal proofs have opposite architectural roles. For Euclid, I.7 is preparation for SSS: the isosceles theorem and angle order exclude a second same-side vertex. In the current reconstruction, SSS is already available and I.7 becomes a concise corollary of stronger reusable uniqueness infrastructure.

The audit corrects the classical dependency by removing I.6, replaces the old formal flowchart by the actual common-ray geometry, and keeps the lower SSS engine modular rather than duplicating the I.8 audit. The result is a three-level account of the same theorem: Euclid’s representative contradiction, the proposition-level Lean proof, and the reusable Hilbert machinery that makes the formal proof short.